

ECO 310: Precept 4

1. Matt has constant absolute risk aversion, with utility $u(w) = 1 - e^{-\alpha w}$ and $\alpha > 0$. He is offered a gamble:

$$\begin{cases} \text{lose } L & \text{w.p. } \frac{1}{2} \\ \text{win } H & \text{w.p. } \frac{1}{2} \end{cases}$$

Find a threshold L^* such that for $L > L^*$, he is unwilling to take the gamble even for infinite H .

Answer: if he accepts the gamble, his expected utility – assuming initial wealth w_0 – is

$$\begin{aligned} & \frac{1}{2}u(w_0 - L) + \frac{1}{2}u(w_0 + H) \\ &= \frac{1}{2} \left(1 - e^{-\alpha(w_0 - L)} \right) + \frac{1}{2} \left(1 - e^{-\alpha(w_0 + H)} \right) \\ &= 1 - \frac{1}{2}e^{-\alpha(w_0 - L)} - \frac{1}{2}e^{-\alpha(w_0 + H)} \end{aligned}$$

If he doesn't, his utility is $1 - e^{-\alpha w_0}$. So he'll reject the gamble if

$$\begin{aligned} 1 - e^{-\alpha w_0} &\geq 1 - \frac{1}{2}e^{-\alpha(w_0 - L)} - \frac{1}{2}e^{-\alpha(w_0 + H)} \\ \iff \frac{1}{2}e^{-\alpha(w_0 - L)} + \frac{1}{2}e^{-\alpha(w_0 + H)} &\geq e^{-\alpha w_0} \\ \iff \frac{1}{2}e^{\alpha L} + \frac{1}{2}e^{-\alpha H} &\geq 1 \end{aligned}$$

As $H \rightarrow \infty$, so $\frac{1}{2}e^{-\alpha H} \rightarrow 0$, this becomes $e^{\alpha L} \geq 2 \iff \alpha L \geq \ln 2 \iff L \geq \frac{\ln 2}{\alpha}$. So the threshold is $\frac{\ln 2}{\alpha}$.

2. A risk-lover with \$400 expects to gain \$200 with chance $\frac{1}{6}$, and lose \$100 with chance $\frac{5}{6}$. Show that this individual will pay at most \$50 to avoid this risk.

Answer #1: the expected wealth change in this game is $\frac{1}{6}(200) + \frac{5}{6}(-100) = -50$. Thus, a risk-neutral person would be willing to pay exactly the expected loss, 50, to avoid the risk. A risk-lover would thus pay at most 50, since they actually enjoy the risk.

Answer #2: mathematically, this person is only willing to pay x to avoid the gamble if

$$u(400 - x) \geq \frac{1}{6}u(400 + 200) + \frac{5}{6}u(400 - 100)$$

(The LHS is utility if you pay x and just keep \$400, and the RHS is the expected utility of playing the game). For a risk-lover, expected utility EXCEEDS utility of expected wealth, so the RHS exceeds

$$u \left(\frac{1}{6}(400 + 200) + \frac{5}{6}(400 - 100) \right) = u \left(100 + \frac{5(300)}{6} = u(350) \right)$$

So, they are only willing to pay x iff

$$u(400 - x) \geq u(350) \Rightarrow 400 - x \geq 350 \Rightarrow x \leq 50$$

3. Imagine that you are an Eagles fan, considering betting on the Superbowl. The moneyline for betting on the Chiefs is -120 , meaning that you must bet \$120 to win (your money back plus) \$100: *equivalently, the return rate of a bet if the Chiefs win is $r = \frac{100}{120} = \frac{5}{6}$* . (Recall from Lecture 9: return rate r means that if you invest m , your wealth changes by rm). You expect the Chiefs to win with probability $p > \frac{1}{2}$.

Write out the equation describing your optimal bet x on the Chiefs, assuming that if they win, it's like a loss of L for you (with an Eagles win normalized to a gain of \$0). You care only about wealth with $u(w) = \ln w$. Explain also why if the odds are unfavorable, namely $\frac{p}{1-p} < \frac{6}{5}$ (this means you expect to lose money, since you gain $\frac{5}{6}x$ with the probability p that they win but lose x with the probability $(1-p)$ that they lose, and here $\frac{5}{6}xp - (1-p)x < 0$), then you'll bet a positive amount on the Chiefs, but a small enough amount that you still prefer they lose.

Answer: Mathematically, let x be the amount you bet on the chiefs. If they win, your bet changes your wealth by $\frac{5}{6}x$, but then winning hurts you by L , so your final wealth is $w_0 - L + \frac{5}{6}x$. If they lose, you lose your bet (so your wealth changes by $-x$), but at least you don't suffer the loss of a team you dislike winning. So wealth is $w_0 - x$. Thus, expected utility from bet x is

$$EU(x) = p \ln \left(w_0 - L + \frac{5}{6}x \right) + (1-p) \ln(w_0 - x)$$

The FOC for an interior x is

$$\frac{5}{6} \frac{p}{w_0 - L + \frac{5}{6}x} - \frac{1-p}{w_0 - x} = 0$$

Rearranging,

$$\frac{5}{6} \frac{p}{1-p} = \frac{w_0 - L + \frac{5}{6}x}{w_0 - x}$$

We can directly see here that if $\frac{p}{1-p} < \frac{6}{5}$, the LHS is smaller than 1, so this is solved via

$$\underbrace{w_0 - L + \frac{5}{6}x}_{\text{wealth if Chiefs win}} < \underbrace{w_0 - x}_{\text{wealth if Eagles win}}$$

Thus, since more money is better, you prefer that the Eagles win.

Intuitively, as an Eagles fan, the Chiefs winning is like an "accident" for you, and so betting on the Chiefs is like insurance against this accident—you get some money back if it happens. A risk averse person typically does want insurance, so that's why betting on the Chiefs beats betting on the Eagles under the assumption that you only care about wealth. But we saw in the insurance problem that with unfavorable odds, you will less than fully insure, meaning you still prefer the accident not happen.